

# ASP3231 Assignment 3

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## Question 1

1a.

Wien's Displacement Law relates the wavelength at which a blackbody is most radiant to the inverse of its temperature. It is defined under the assumption that the object is a perfectly radiating blackbody, which is a typical assumption for analysing most stars. It follows that

$$\lambda_{\max} = \frac{2.898 \cdot 10^{-3} \text{m} \cdot \text{K}}{T}. \quad (1)$$

On the  $t=1.43$  days graph in Figure 1, the wavelength that is emitted at the highest flux (intensity) is approximately  $1.2 \cdot 10^3$  angstroms, or  $1.2 \cdot 10^{-7}$  metres. Hence, the temperature of the object 1.43 days after the explosion is  $\approx 2.415 \cdot 10^4$  K.

1b.

Wien's Displacement Law essentially tells us that an increase in the wavelength (lengthening) which corresponds to the brightest flux leads to a decrease in temperature. We can see that over time, the maximum wavelength is getting longer while the maximum intensity decreases, which means the temperature is decreasing over time. Using the same approximation as in Q1a, the temperature between the first and last epoch has decreased by approximately  $1.84 \cdot 10^4$  K, where the temperature after 24.01 days is  $\approx 5796$  K.

1c.

We start to observe a significant deviation from the blackbody fit around the 14.94-day epoch. Beyond this there is a very significant deviation. This implies that the blackbody fit is no longer a reliable tool to estimate the temperature of the body.

In a more physical sense, the breaking down of the blackbody assumption implies that the supernova is losing opacity as it expands and cools. We start to lose the continuum emission of a perfect blackbody and see the individual emission lines start to dominate as the supernova becomes optically thin.

1d.

The flux at  $10^4$  angstroms of the object over time actually remains fairly constant. We can see that the last few data points, and even the blackbody fit, for the most part remain close to the  $10^{-17}$  flux unit mark. The only noticeable shift is a slight increase at first followed by a slight decrease later.

For a blackbody, flux is directly proportional to temperature and the solid angle. Assuming the flux remains mostly constant, and the temperature is decreasing, the solid angle must be increasing over time. This physically represents the expansion of the supernova. Finding an exact size would require knowledge of the distance to the supernova.

1e.

The proportionality described in Q1d is known as the Rayleigh-Jeans approximation and is given by

$$B_\nu = . \quad (2)$$

## Question 2

2a.

2b.

2c.

2d.